

Question:

Find the sum of all integer bases $b > 9$ for which 17_b is a divisor of 97_b .

Reasoning + Answer:

<think>

1. Understand the problem :

- We are given a number in base b , denoted as 17_b .
- We need to find all integer bases $b > 9$ such that 17_b is a divisor of 97_b .
- Convert 17_b and 97_b to decimal .
- Check which bases satisfy the divisibility condition .

2. Convert the numbers to decimal :

- 17_b in decimal is $1 \cdot b + 7$.
- 97_b in decimal is $9 \cdot b + 7$.

3. Set up the divisibility condition :

- 17_b is a divisor of 97_b , i.e., $(1 \cdot b + 7)$ is a divisor of $(9 \cdot b + 7)$.

4. Factorize the divisibility condition :

- Let $1 \cdot b + 7 = c$ and $9 \cdot b + 7 = d$.
- Then , d must be divisible by c .
- $d = 9b + 7$ and $c = b + 7$.

5. Solve the divisibility condition :

- $9b + 7$ is divisible by $b + 7$.
- Substitute $b + 7$ with c :
- $9b + 7 = k \cdot (b + 7)$ for some integer k .
- Solve for b :
- $9b + 7 = k(b + 7)$.
- $9b + 7 = kb + 7k$.
- Rearrange : $9b - kb = 7k - 7$.
- Factor : $b(9 - k) = 7(k - 1)$.
- $b = \frac{7(k - 1)}{9 - k}$.

6. Find integer solutions for b :

- b must be an integer , so $\frac{7(k - 1)}{9 - k}$ must be an integer .
- $9 - k$ must be a factor of 7 .
- The factors of 7 are $\pm 1, \pm 7$.
- Solve for k :
- $9 - k = 7 \Rightarrow k = 2 \Rightarrow b = 7 / 1 = 7$ (not valid since $b > 9$).
- $9 - k = -1 \Rightarrow k = 8 \Rightarrow b = 7 / 8$ (not an integer).
- $9 - k = -7 \Rightarrow k = 16 \Rightarrow b = 7 / 16$ (not an integer).
- $9 - k = 1 \Rightarrow k = 8 \Rightarrow b = 7 / 8$ (not valid).
- $9 - k = 1 \Rightarrow k = 8 \Rightarrow b = 7 / 8$ (not an integer).

7. Find valid integer solutions :

- For $b = 16$ (base greater than 9):
- Check $b + 7 = 16 + 7 = 23$:
- Check $9b + 7 = 9(16) + 7 = 144 + 7 = 151$